

TWO ZONOTOPES WITH THE SAME ARITHMETIC MATROID AND DIFFERENT GRADED EHRHART SERIES

ABSTRACT. Crowley and Partida ask whether the graded Ehrhart theory of a non-unimodular zonotope is an invariant of its associated arithmetic matroid. The answer is no, by a pair of lattice parallelograms in $\mathbb{Z}^2$: the zonotopes Z_1 generated by $(1,0), (1,5)$ and Z_2 generated by $(1,0), (2,5)$ have literally equal arithmetic matroids, hence the same arithmetic Tutte polynomial $x^2 + 4$ and the same ungraded lattice point counts $5m^2 + 2m + 1$, yet

$$i_{Z_1}(1; q) = 1 + 2q + 2q^2 + q^3 + q^4 + q^5, \quad i_{Z_2}(1; q) = 1 + 2q + 3q^2 + 2q^3.$$

The witness is exhibited in full and the separation is proved by hand at $m = 1$.

1. THE QUESTION

Let $S \subset \mathbb{Z}^d$ be finite, let $I(S) \subseteq \mathbb{C}[x_1, \dots, x_d]$ be its vanishing ideal, and let $\text{gr } I(S)$ be the homogeneous ideal generated by the top-degree homogeneous components of all elements of $I(S)$. Put

$$\text{Orb}(S) = \mathbb{C}[x_1, \dots, x_d] / \text{gr } I(S),$$

a graded ring of total dimension $|S|$. For a lattice polytope $P \subseteq \mathbb{R}^d$ and an integer $m \geq 0$, the *graded Ehrhart series* of P is

$$(1) \quad i_P(m; q) = \text{Hilb}(\text{Orb}(mP \cap \mathbb{Z}^d); q),$$

so that $i_P(m; 1) = |mP \cap \mathbb{Z}^d|$ is the ordinary Ehrhart polynomial. This is the notion of Reiner and Rhoades [3]; (1) is Definition 2.4 of Crowley and Partida [1].

Let $v_1, \dots, v_n \in \mathbb{Z}^d$ and let $Z = Z(v_1, \dots, v_n) = \{\sum_i t_i v_i : 0 \leq t_i \leq 1\}$ be the zonotope they generate. Its *arithmetic matroid* is the represented matroid on $E = \{1, \dots, n\}$ together with the *multiplicity function*

$$(2) \quad \mu(S) = [(\text{span}_{\mathbb{R}}\{v_i : i \in S\} \cap \mathbb{Z}^d) : \sum_{i \in S} \mathbb{Z}v_i],$$

the index of the lattice generated by S inside the saturation of its span; on a basis this is $|\det|$. D’Adderio and Moci [2] proved that the ungraded counts are determined by these data: $|mZ \cap \mathbb{Z}^d| = m^d T_M\left(\frac{m+1}{m}, 1\right)$, where T_M is the arithmetic Tutte polynomial. Crowley and Partida prove that for a *unimodular* zonotope the graded series is likewise matroidal, and then ask:

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Question (Crowley–Partida [1]). Is the graded Ehrhart theory of a non-unimodular zonotope Z an invariant of its associated arithmetic matroid? If so, is there a generalization of [their proposition on Ehrhart polynomials] to arithmetic matroids?

The wording is quoted from the source of [1], where the question stands in the subsection *The graded Ehrhart theory of non-unimodular zonotopes*. Only the first clause is at issue here. The same subsection states the expectation:

“[...] it is still possible that the q -lattice point counts of an arbitrary zonotope are an arithmetic matroid invariant. It would be quite surprising to find two zonotopes Z and Z' whose arithmetic matroids are the same but $i_Z(m; q) \neq i_{Z'}(m; q)$ for some integer $m \geq 0$.”

2. THE WITNESS

Theorem 1. *Let*

$$A_1 = \begin{pmatrix} 1 & 1 \\ 0 & 5 \end{pmatrix}, \quad A_2 = \begin{pmatrix} 1 & 2 \\ 0 & 5 \end{pmatrix},$$

and let $Z_1 = Z((1,0), (1,5))$ and $Z_2 = Z((1,0), (2,5))$ be the zonotopes generated by their columns. Then:

- (a) *the arithmetic matroids of Z_1 and Z_2 on the ground set $E = \{1, 2\}$ are equal, not merely isomorphic: both are $U_{2,2}$ with $\mu(\emptyset) = \mu(\{1\}) = \mu(\{2\}) = 1$ and $\mu(\{1, 2\}) = 5$, hence both have arithmetic Tutte polynomial $T_M(x, y) = x^2 + 4$ and, for every $m \geq 0$, $|mZ_1 \cap \mathbb{Z}^2| = |mZ_2 \cap \mathbb{Z}^2| = 5m^2 + 2m + 1$;*
- (b)

$$i_{Z_1}(1; q) = 1 + 2q + 2q^2 + q^3 + q^4 + q^5 \neq 1 + 2q + 3q^2 + 2q^3 = i_{Z_2}(1; q).$$

Consequently the answer to the first clause of the question above is **no**.

The two lattice point sets at $m = 1$ are, in full,

$$(3) \quad S_1 = Z_1 \cap \mathbb{Z}^2 = \{(0,0), (1,0), (1,1), (1,2), (1,3), (1,4), (1,5), (2,5)\},$$

$$(4) \quad S_2 = Z_2 \cap \mathbb{Z}^2 = \{(0,0), (1,0), (1,1), (1,2), (2,3), (2,4), (2,5), (3,5)\}.$$

Both have eight elements. Six of the eight points of S_1 are collinear; the largest collinear subset of S_2 has three points. That difference in configuration is the whole phenomenon, and Section 3 turns it into a proof.

Part (a) is immediate: all four generating columns are primitive, so $\mu(\{1\}) = \mu(\{2\}) = 1$ for both, and $|\det A_1| = |\det A_2| = 5$, so $\mu(\{1, 2\}) = 5$ for both. Hence $T_M(x, y) = (x - 1)^2 + 2(x - 1) + 5 = x^2 + 4$, and the D’Adderio–Moci identity gives $m^2(((m + 1)/m)^2 + 4) = 5m^2 + 2m + 1$ for both. (Equivalently, by Pick’s theorem: area $5m^2$, boundary points $B = 4m$ because all four edges are primitive, interior points $I = 5m^2 - 2m + 1$.)

3. PROOF OF THEOREM 1(B)

For a finite $S \subset \mathbb{Z}^2$ and $j \geq 0$ let $r_j(S)$ be the rank of the evaluation matrix of the monomials of degree at most j on S , and let $N_j = \binom{j+2}{2}$ be the number of those monomials. Write W_j for the space of degree- j homogeneous top components of elements of $I(S)$ of degree at most j .

Lemma 2. $\dim \text{Orb}(S)_j = r_j(S) - r_{j-1}(S)$ for all $j \geq 0$, with $r_{-1} = 0$.

Proof. Taking the degree- j component is a linear surjection from $\{f \in I(S) : \deg f \leq j\}$ onto W_j with kernel $\{f \in I(S) : \deg f \leq j-1\}$, so $\dim W_j = (N_j - r_j) - (N_{j-1} - r_{j-1}) = (j+1) - (r_j - r_{j-1})$. Moreover $xW_{j-1} + yW_{j-1} \subseteq W_j$: if $f \in I(S)$ has degree $\leq j-1$ then $xf \in I(S)$ has degree $\leq j$ and its degree- j component is x times the degree- $(j-1)$ component of f . Hence the homogeneous ideal generated by $\bigcup_i W_i$ has degree- j part exactly W_j , that is $(\text{gr } I(S))_j = W_j$, whence $\dim \text{Orb}(S)_j = \dim \mathbb{C}[x, y]_j - \dim W_j = (j+1) - \dim W_j = r_j - r_{j-1}$. $\square$

S_1 : a forced line component. S_1 from (3) has six points on the line $x = 1$, namely $(1, y)$ for $0 \leq y \leq 5$, and the two extra points $(0, 0)$ and $(2, 5)$. Let $2 \leq j \leq 5$ and let f have degree at most j and vanish on S_1 . Restricting f to $x = 1$ gives a polynomial in y of degree at most $j \leq 5$ with six distinct roots, hence identically zero; therefore $(x-1) \mid f$. Since $x-1$ is nonzero at $(0, 0)$ and at $(2, 5)$, the cofactor $g = f/(x-1)$ has degree at most $j-1$ and vanishes at those two points, and the conditions $g(0, 0) = g(2, 5) = 0$ are independent on $\mathbb{C}[x, y]_{\leq j-1}$ for $j \geq 2$ because 1 and x separate the two points. Conversely every such product vanishes on S_1 . Hence

$$\begin{aligned} \dim\{f \in I(S_1) : \deg f \leq j\} &= N_{j-1} - 2, \\ r_j(S_1) &= N_j - N_{j-1} + 2 = j + 3 \quad (2 \leq j \leq 5). \end{aligned}$$

With $r_0 = 1$ and $r_1 = 3$ the profile is $r_\bullet = (1, 3, 5, 6, 7, 8)$, and Lemma 2 gives $\dim \text{Orb}(S_1)_\bullet = (1, 2, 2, 1, 1, 1)$, i.e. $i_{Z_1}(1; q) = 1 + 2q + 2q^2 + q^3 + q^4 + q^5$. The coefficients sum to $8 = |S_1|$.

S_2 : no conic, and eight separating cubics. Label the points of (4)

$$\begin{aligned} P_1 &= (0, 0), & P_2 &= (1, 0), & P_3 &= (1, 1), & P_4 &= (1, 2), \\ P_5 &= (2, 3), & P_6 &= (2, 4), & P_7 &= (2, 5), & P_8 &= (3, 5). \end{aligned}$$

Clearly $r_0 = 1$ and $r_1 = 3$ (the points are not collinear).

$r_2 = 6$, that is, no conic passes through S_2 . A conic meets a line in at most two points unless the line is one of its components. The three points P_2, P_3, P_4 lie on $x = 1$ and the three points P_5, P_6, P_7 lie on $x = 2$, so a conic through S_2 must be a scalar multiple of $(x-1)(x-2)$; that polynomial takes the value $(-1)(-2) = 2$ at $P_1 = (0, 0)$, a contradiction. Since $N_2 = 6$, the eight points impose 6 independent conditions on conics.

$r_3 = 8$, that is, the eight points impose independent conditions on cubics. It suffices to exhibit, for each i , a cubic vanishing at the seven points P_j ,

$j \neq i$, and not at P_i . Table 1 does so; each cubic is a product of three lines, and the column “lines cover” shows how the seven points are distributed among them, so the vanishing is visible without any substitution. The stated value at the omitted point is the product of the three displayed linear forms evaluated there.

omit	cubic	value	lines cover
$P_1 = (0, 0)$	$(x - 1)(x - 2)(x - 3)$	-6	$x=1$: $P_2P_3P_4$; $x=2$: $P_5P_6P_7$; $x=3$: P_8
$P_2 = (1, 0)$	$(y - 2x)(x - 2)(y - 2x + 1)$	-2	$y=2x$: $P_1P_4P_6$; $x=2$: P_5P_7 ; $y=2x-1$: P_3P_8
$P_3 = (1, 1)$	$y(x - 2)(2y - 3x - 1)$	2	$y=0$: P_1P_2 ; $x=2$: $P_5P_6P_7$; $2y=3x+1$: P_4P_8
$P_4 = (1, 2)$	$y(x - 2)(y - 2x + 1)$	-2	$y=0$: P_1P_2 ; $x=2$: $P_5P_6P_7$; $y=2x-1$: P_3P_8
$P_5 = (2, 3)$	$(y - 5)(x - 1)(y - 2x)$	2	$y=5$: P_7P_8 ; $x=1$: $P_2P_3P_4$; $y=2x$: P_1P_6
$P_6 = (2, 4)$	$(y - 5)(x - 1)(3x - 2y)$	2	$y=5$: P_7P_8 ; $x=1$: $P_2P_3P_4$; $3x=2y$: P_1P_5
$P_7 = (2, 5)$	$(y - 2x)(x - 1)(y - 2x + 1)$	2	$y=2x$: $P_1P_4P_6$; $x=1$: P_2P_3 ; $y=2x-1$: P_5P_8
$P_8 = (3, 5)$	$x(x - 1)(x - 2)$	6	$x=0$: P_1 ; $x=1$: $P_2P_3P_4$; $x=2$: $P_5P_6P_7$

TABLE 1. Eight cubics separating the points of S_2 , proving $r_3(S_2) = 8$.

Hence $r_\bullet(S_2) = (1, 3, 6, 8)$ and $r_3 = 8 = |S_2|$, so by Lemma 2 $\dim \text{Orb}(S_2)_\bullet = (1, 2, 3, 2)$ and $i_{Z_2}(1; q) = 1 + 2q + 3q^2 + 2q^3$, again of coefficient sum 8. The two series differ already in the coefficient of q^2 (2 against 3) and in degree (5 against 3), while part (a) makes the arithmetic matroids equal. One value of m suffices, so Theorem 1 is proved. $\square$

The graded series are compared here only at $m = 1$; the second clause of the question and the case of three or more generators are not addressed.

4. REPRODUCTION AND VERIFICATION

Everything above can be redone by hand from (3), (4) and Section 3; no computation is needed to check Theorem 1.

A program `verify.py` accompanies this note, together with a recorded run `verify.output.txt`. It re-derives every numerical claim above from the two generator matrices alone, in exact integer and rational arithmetic: the multiplicity function by integer Smith normal form, the lattice point sets by two independent tests cross-checked against (3), (4) and Pick’s theorem, and each graded series both by the rank differencing of Lemma 2 and from

an explicit basis of $\{f \in I(S) : \deg f \leq j\}$ and its degree- j top forms. It confirms every row of Table 1, including the printed value of each cubic at its omitted point. All 48 of its checks pass; several of them concern material not used above, and its output lists what it does not cover, the bibliographic facts among them: those are read from the sources, not computed.

REFERENCES

- [1] S. Crowley and D. Partida, *Graded Ehrhart theory of unimodular zonotopes*, arXiv:2603.07873, 2026. The question quoted in Section 1 is Question 7.3 there, read from the L^AT_EX source of the e-print.
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